mission computer enables it to perform independent reconnaissance and target engagement with minimal operator input.

Impact: The Ghatak's successful prototype trials marked India's entry into the stealth UCAV domain, strengthening its defense self-reliance and positioning it among nations with advanced unmanned combat capabilities.

5.9 Summary

The fourth generation of drones marks a pivotal advancement in unmanned aerial technology, defined by the seamless integration of artificial intelligence (AI), autonomous operation, stealth engineering, and networked warfare systems. These drones possess the ability to perform complex missions independently through AI-driven decision-making, sensor fusion, and real-time data processing, reducing human intervention while enhancing precision and adaptability. Evolving from earlier generations, fourth-generation UAVs introduced innovations such as swarm intelligence, manned–unmanned teaming (MUM-T), and AI-enabled mission coordination, making them indispensable for both military and civilian applications – from precision strikes, surveillance, and electronic warfare to disaster relief, environmental monitoring, and communication support. Notable systems, including the XQ-58A Valkyrie, CH-7 Rainbow, Skyborg, Boeing ATS, FCAS, Tengden TW-356, Ghatak UCAV, and Okhotnik-B, exemplify this evolution toward intelligent, modular, and stealth-optimized aerial platforms. Despite their revolutionary impact, challenges persist in cybersecurity, operational reliability, ethical regulation, and cost efficiency, underscoring the need for global governance and responsible AI deployment. Through real-world case studies ranging from autonomous combat integration to post-disaster swarm operations, this chapter underscores how fourth-generation drones have transformed airpower into a collaborative, intelligent, and multi-domain system, setting